

ast paper 2022:

Answer the short questions:

(i) What are the problems of estimation? Explain each.

Estimation is a critical aspect of project management, particularly in fields like software development, construction, and engineering. However, it is fraught with several challenges and problems, which can affect the accuracy and reliability of the estimates. Here are some common problems of estimation.

1: Uncertainty and incomplete information:

Lack of full project details leads to inaccurate estimates.

2: Lack of Experience:

Inexperienced estimators may overlook important factors.

3: Scope creep:

Adding extra features without adjusting estimates results in overruns.

4: Over-optimism:

Unrealistic expectations about time and resources lead to tight schedules.

5: Underestimation of Risks:

Ignoring potential risks causes unexpected delays and costs.

(ii)

What is the importance of sufficiency and joint sufficiency as a property of good estimator?

Importance of sufficiency:

Complete information: A sufficient statistics captures all the information about a parameter.

Efficiency: leads to more reliable and precise estimator
Simplification: Reduced data to a single statistics without losing information. compare efficient

Importance of joint sufficiency:

Combined information:

Multiple statistics together provide all the information about a parameter.

Improved estimation:

Lead to better and more accurate estimates.

Flexibility:

Uses multiple statistics to capture different aspects of the data.

(iii)
The Cramer-Rao inequality is basis of efficiency; discuss briefly?

Cramer Rao lower bound:

It states that the variance of any unbiased estimator is atleast as large as the inverse of the fisher information.

In Mathematically if $\hat{\theta}$ is an unbiased estimator of a parameters then

$$\text{Var}(\hat{\theta}) \geq \frac{1}{I(\theta)}$$

Where $I(\theta)$ is the fisher information.

(iv)

Compare best asymptotically normal estimator with efficient estimator?

Best Asymptotically Normal estimator

• Meaning:

A sample size gets really big, the estimator's distribution looks like a normal distribution.

• Focus:

It becomes approximately normal which simplifies inference.

Efficient estimator:

• Meaning

The estimator achieves the smallest possible variance among all unbiased estimators.

Focus

It reaches the Cramer-Rao lower bound, making it the most precise and reliable.

Efficient estimators are always BAN, but BAN estimators are not always ~~same~~ efficient.

(v)

Differentiate between location and scale invariant

Estimators:

Location invariant Estimators and scale invariant Estimators are two types of estimators that remain unaffected by certain transformations of the data.

Location invariant estimators

An estimator is location invariant if it does not change when a constant is added to all observations.

Example

If $x_1, x_2, \dots, x_n$ are observations and $y_i = c x_i$ for some constant c , a location invariant estimator $\hat{\theta}$ satisfies:

$$\hat{\theta}(y_1, \dots, y_n) = \hat{\theta}(x_1, \dots, x_n)$$

2. Scale Invariant Estimators

An estimator is scale invariant if it remains unchanged when all observations are multiplied by a positive constant.

Example

If $x_1, x_2, \dots, x_n$ are observations and $y_i = c x_i$ for some positive constant c , a scale invariant estimator of $\hat{\theta}$ satisfies:

$$\hat{\theta}(y_1, \dots, y_n) = \hat{\theta}(x_1, \dots, x_n) \quad (vi)$$

How the property of unbiasedness and efficiency are related? Which one is more needed?

• Unbiasedness:

The estimator's expected value equals the true parameter. Ensures no systematic error.

Efficiency:

The estimator achieves the minimum variance among all unbiased estimators, meeting the Cramer-Rao lower bound.

Relationship:

Efficiency requires unbiasedness, but not all unbiased estimators are efficient.

which is more needed:

Unbiasedness is crucial for correctness.

Efficiency is key for precision, especially in large samples.

Long Questions

Answer the following questions.

- (i) Let $t = x_1 + x_2$ $k = x_1 + 2x_2$ be two statistics where x_i follow poisson distribution, check t and k for possible sufficiency.

The probability Mass function of poisson dist is

$$f(x, \lambda) = \frac{e^{-\lambda} \lambda^x}{x!}, x > 0$$

The joint probability function for x_1, x_2

$$f(x_1, x_2, \lambda) = f(x_1; \lambda) f(x_2; \lambda) = \frac{e^{-\lambda} \lambda^{x_1}}{x_1!} \frac{e^{-\lambda} \lambda^{x_2}}{x_2!} = e^{-2\lambda} \frac{\lambda^{x_1+x_2}}{x_1! x_2!} \quad \text{--- (A)}$$

Consider

$$T = x_1 + x_2$$

$$\begin{aligned} E(T) &= E(x_1) + E(x_2) \\ &= \lambda + \lambda \\ &= 2\lambda \end{aligned}$$

The sum of poisson random variable is also poisson random variables, so

$$f(T, 2\lambda) = \frac{e^{-2\lambda} (2\lambda)^T}{T!}$$

Now, we find the conditional mass function of x_1 and x_2 given T_1

$$f(x_1, x_2 | T_1 = x_1 + x_2) = \frac{f(x_1, x_2; \lambda)}{f(T_1)}$$

$$f(x_1, x_2 | T_1 = x_1 + x_2) = \left(\frac{e^{-2\lambda} \lambda^{x_1+x_2}}{x_1! x_2!} \right) \left(\frac{T_1!}{e^{-2\lambda} (2\lambda)^{T_1}} \right)$$

$$f(x_1, x_2 | T_1 = x_1 + x_2) = \left(\frac{\lambda^{x_1+x_2}}{x_1! x_2!} \right) \left(\frac{T_1!}{2^{T_1} \lambda^{x_1+x_2}} \right) = \left(\frac{T_1!}{2^{T_1} x_1! x_2!} \right)$$

The conditional distribution is independent of parameter, hence T is sufficient statistics

Now take,

$$K = X_1 + 2X_2$$

$$E(K) = E(X_1) + 2E(X_2)$$

$$\lambda + 2\lambda = 3\lambda$$

The probability Mass function of K

$$f(K; 3\lambda) = \frac{e^{-3\lambda} (3\lambda)^K}{K!}$$

Now we find the conditional Mass function of X_1 and X_2 given K .

$$f(X_1, X_2 | K = x_1 + 2x_2) = \frac{f(x_1, x_2; \lambda)}{f(K)}$$

$$f(X_1, X_2 | K = x_1 + 2x_2) = \left(\frac{e^{-2\lambda} \lambda^{x_1+x_2}}{x_1! x_2!} \right) \left(\frac{K!}{e^{-3\lambda} (3\lambda)^K} \right)$$

$$f(X_1, X_2 | K = x_1 + 2x_2) = \left(\frac{e^{-2\lambda+3\lambda} \lambda^{x_1+x_2-x_1-2x_2}}{x_1! x_2!} \right) \left(\frac{K!}{(3)^K} \right)$$

$$f(X_1, X_2 | K = x_1 + 2x_2) = \left(\frac{e^{\lambda} \lambda^{-x_2}}{x_1! x_2!} \right) \left(\frac{K!}{(3)^K} \right)$$

With reference to Cramer-Rao inequality and with usual notations, prove that

$$(a) E\left(\frac{\partial \ln L}{\partial \theta}\right) = 0$$

$$(b) E\left(\frac{\partial \ln L}{\partial \theta}\right)^2 = -E\left[\frac{\partial^2 \ln L}{\partial \theta^2}\right]$$

Proof:-

Given that

$$L(x_1, x_2, \dots, x_n; \theta) = f(x_1; \theta) \cdot f(x_2; \theta) \cdots f(x_n; \theta)$$

Let

$$L(x_1, x_2, \dots, x_n; \theta) = L \quad \text{Then}$$

$$\int \cdots \int L dx_1 dx_2 \cdots dx_n = 1 \quad (1)$$

Suppose that first two derivatives of L with respect to θ exists $\forall \theta$.

differentiate both sides of eq (1), we get

$$\int \cdots \int \frac{\partial L}{\partial \theta} dx_1 dx_2 \cdots dx_n = 0$$

$$\int \cdots \int \left(\frac{1}{L} \frac{\partial L}{\partial \theta}\right) L dx_1 dx_2 \cdots dx_n = 0$$

$$\int \cdots \int \left(\frac{\partial \ln L}{\partial \theta}\right) L dx_1 dx_2 \cdots dx_n = 0$$

It can be written as

$$E\left[\frac{\partial \ln L}{\partial \theta}\right] = 0 \quad \text{proved}$$

Now,

Let us differentiate eq (2), again we have

$$\iint \dots \int \left\{ \left(\frac{1}{L} \frac{\partial L}{\partial \theta} \right) \left(\frac{\partial L}{\partial \theta} \right) + L \frac{\partial}{\partial \theta} \left(\frac{1}{L} \frac{\partial L}{\partial \theta} \right) \right\} dx_1 dx_2 \dots dx_n = 0$$

$$\iint \dots \int \left\{ \left(\frac{1}{L^2} \frac{\partial L}{\partial \theta} \right) \left(\frac{\partial L}{\partial \theta} \right) L + L \frac{\partial}{\partial \theta} \left(\frac{1}{L} \frac{\partial L}{\partial \theta} \right) \right\} dx_1 dx_2 \dots dx_n = 0$$

$$\iint \dots \int \left\{ \left(\frac{1}{L} \frac{\partial L}{\partial \theta} \right)^2 L + L \frac{\partial}{\partial \theta} \left(\frac{\partial \ln L}{\partial \theta} \right) \right\} dx_1 dx_2 \dots dx_n = 0$$

$$\iint \dots \int \left\{ \left(\frac{\partial \ln L}{\partial \theta} \right)^2 L + L \left(\frac{\partial^2 \ln L}{\partial \theta^2} \right) \right\} dx_1 dx_2 \dots dx_n = 0$$

$$\iint \dots \int \left\{ \left(\frac{\partial \ln L}{\partial \theta} \right)^2 + \left(\frac{\partial^2 \ln L}{\partial \theta^2} \right) \right\} L dx_1 dx_2 \dots dx_n = 0$$

$$\iint \dots \int \left(\frac{\partial \ln L}{\partial \theta} \right)^2 L dx_1 dx_2 \dots dx_n + \iint \dots \int \left(\frac{\partial^2 \ln L}{\partial \theta^2} \right) L dx_1 dx_2 \dots dx_n = 0$$

These can be written as

$$E \left[\frac{\partial \ln L}{\partial \theta} \right]^2 + E \left[\frac{\partial^2 \ln L}{\partial \theta^2} \right] = 0$$

$$E \left[\frac{\partial \ln L}{\partial \theta} \right]^2 = -E \left[\frac{\partial^2 \ln L}{\partial \theta^2} \right] \text{ Proved}$$

(iii) State and prove Lehmann-Scheffe theorem.

Let $x_1, \dots, x_n$ be a random sample from the density $f(x, \theta)$. If $s = s(x_1, \dots, x_n)$ is a complete sufficient statistics and if $T^* = t^*(s)$ a function of s , is an unbiased estimator of $g(\theta)$, Then T^* is an uniformly variance unbiased estimator (UMVUE) of $g(\theta)$.

$\frac{d}{dx} \ln p(x; \theta) = 0$

theorem says that

If a complete sufficient statistics exists and there is an unbiased estimator for $g(\theta)$, then there is a UMVUE for $g(\theta)$.

• The UMVUE is the unique unbiased estimator of $g(\theta)$ which is a function of S .

